Student Name: _____

Student ID# _____

	Check which tutorial for pick-up
T03 Tuesdays 2:30 – 4:20	
T01 Fridays 9:30 – 11:20	

CHEMICAL ENGINEERING 2O04
Fluid Mechanics -Midterm #2

DAY CLASS
Duration 120 minutes (6:00pm – 8:00pm) **D.R. Latulippe**
McMaster University Term Test **Friday March 22nd, 2019**

The test includes **5 questions and 1 bonus question**; ensure your copy of the test is complete.

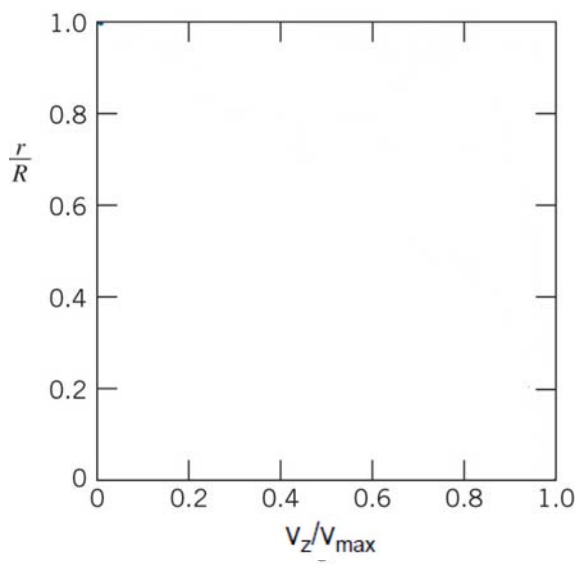
SPECIAL INSTRUCTIONS

1. The class **textbook (any version of the Pritchard's <u>Fox & McDonald</u>) and any notes** are allowed.
2. McMaster calculator only (and case with unit conversions) is permitted.
3. Only tests written in pen will be considered if re-marking is requested (addition errors excepted).
4. On the last two pages, I have posted some useful tables from the lecture notes.
5. Marks for each problem are given (**Total of 50 marks)**. Use this as a time-management guide; for example, a question worth only 2 marks should take less than 5 minutes:

$$2marks\left(\frac{120\ minutes}{50\ marks}\right) = 4.8\ minutes$$

QUESTION 1 - 11 MARKS

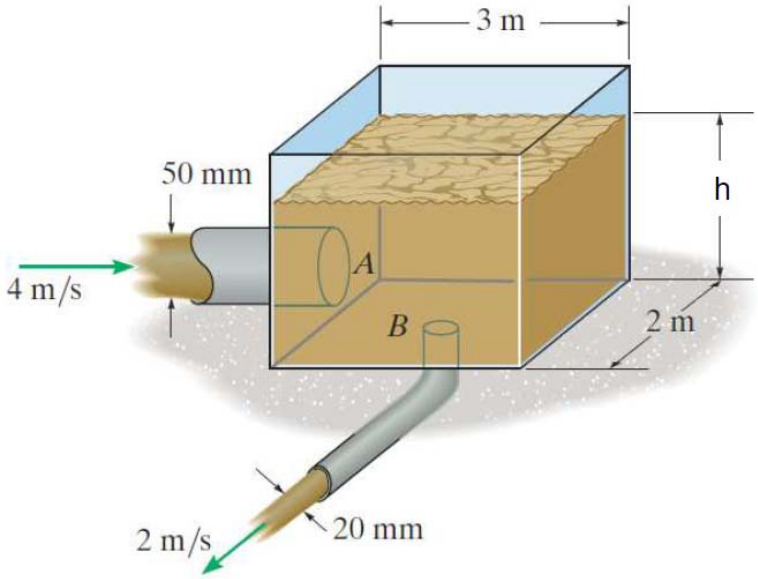
Answer the following short questions in the space provided:

a.	In the figure shown to the right, sketch the normalized velocity profiles in the axial direction (i.e. v_z/v_{max}) as a function of normalized radial position (i.e. r/R) for both laminar flow and turbulent flow conditions. **(2 marks)**	
b.	List the two specific conditions that are required to correctly use the Blasius empirical expression to calculate the friction factor. **(2 marks)**	

c.	In deriving the Bernoulli equation, we defined a term called 'rate of flow work' with SI units of J/s. Starting from the equation that relates work to force and distance, derive an equation for the 'rate of flow work' in terms of pressure, density, and mass flow rate. **(2 marks)**	
d.	State two strategies you can use to prevent cavitation in a piping system (note: 'increasing NPSHA' is not a sufficiently detailed answer) **(2 marks)**	
e.	Table 8.1 lists the roughness of common pipe materials. Do you expect these values to increase OR decrease with pipe age? **(1 mark)**	
f.	Prove that for flow through a square channel with sides of length s, the hydraulic diameter (D_h) is equal to the side length s. **(2 marks)**	

QUESTION 2 - 5 MARKS

Oil (with specific gravity = 0.76) is flowing into the rectangular tank with a constant average velocity of 4 m/s through the 50 mm diameter pipe at A. It flows out of the tank at a constant average velocity of 2 m/s through the 20 mm diameter pipe at B.

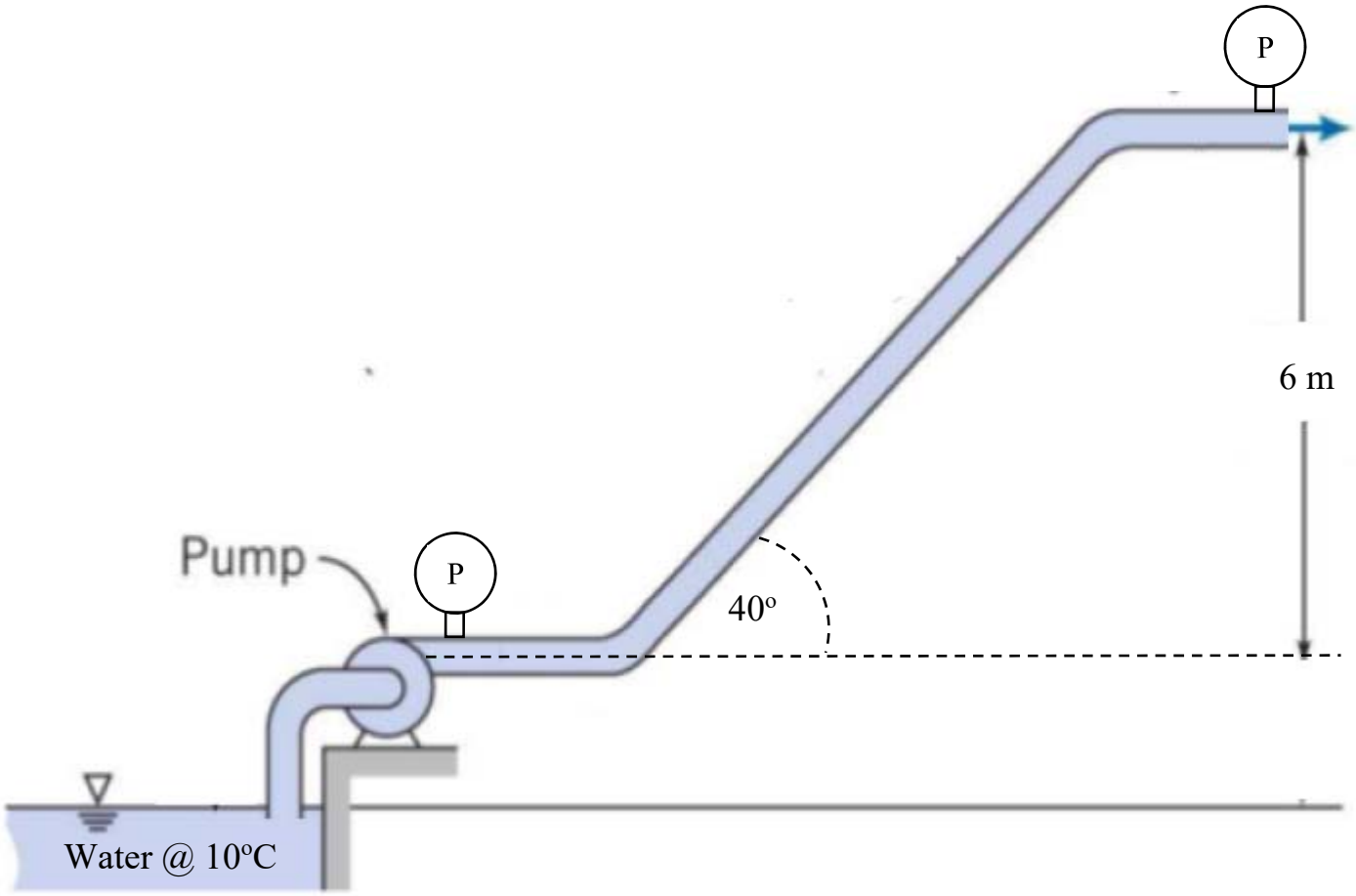


a. Selecting the control volume to include the entire tank contents, write the general form of the mass balance equation for the oil in the tank. **(2 marks)**

b. If the tank is initially empty, determine the time (mins) it takes to fill the tank to a height of 1.5 m **(3 marks)**

QUESTION 3 - 10 MARKS

A new pump-piping system is to be designed to deliver 500 L/min of water (at 10 ºC) uphill from a large reservoir to a construction site. As shown in the figure below, the upper portion of the piping system is 6 m above the lower portion and the angle of the inclined section of pipe is 40º relative to the horizon. The same type of piping (2½" schedule 40 *cast iron*) is used for the entire length on the discharge side of the pump.



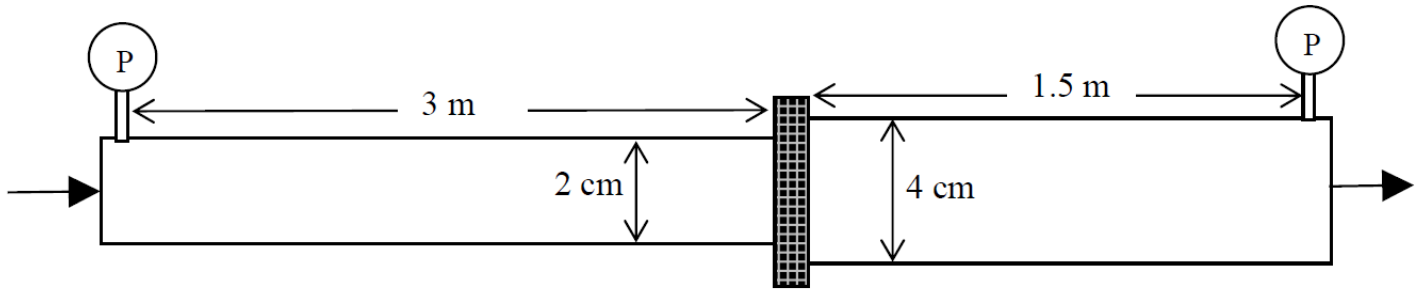
a. Write the simplified form of the Bernoulli equation for the condition where reference points 1 & 2 are chosen to be where the two pressure gauges are located on the discharge side of the pump. NOTE: you can ignore the minor losses associated with the two elbows that are needed to make the incline. **(3 marks)**

b. If the differential pressure reading between the two pressure gauges is reported to be 84.5 kPa, determine the total length of the 2½" schedule 40 *cast iron* piping between the two pressure gauges. **(5 marks)**

c. With extended use, it is expected that the cast iron pipe will become worn and rusty. Comment on which variable in the Bernoulli equation would account for this change and what is the overall effect on the performance of the pump-piping system. Give a BRIEF explanation (no calculations are needed!) of how you would determine the actual change in the performance of the pump-piping system. **(2 marks)**

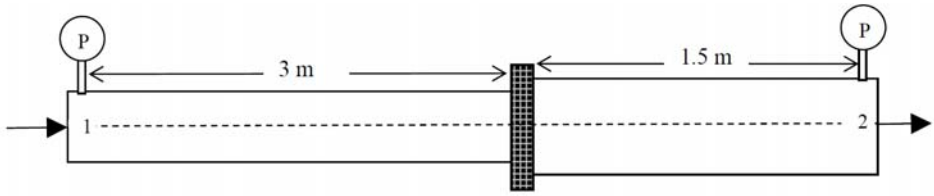
QUESTION 4 - 14 MARKS

Carbon tetrachloride at ambient temperature (density = 1595 kg/m^3; viscosity = 0.001 Pa·s) is flowing from left to right through the horizontal piping system as shown below (not drawn to scale). An expansion fitting is used to connect the smaller (i.e. 2 cm internal diameter) *wrought iron* pipe to the larger (i.e. 4 cm internal diameter) *galvanized iron* pipe; the corresponding loss coefficient K for the expansion fitting is 0.55 with a characteristic velocity corresponding to the velocity in the smaller wrought iron pipe.



a. Using a steady-state mass balance, write the appropriate equation that relates the average velocities in the two pipes.
(2 marks)

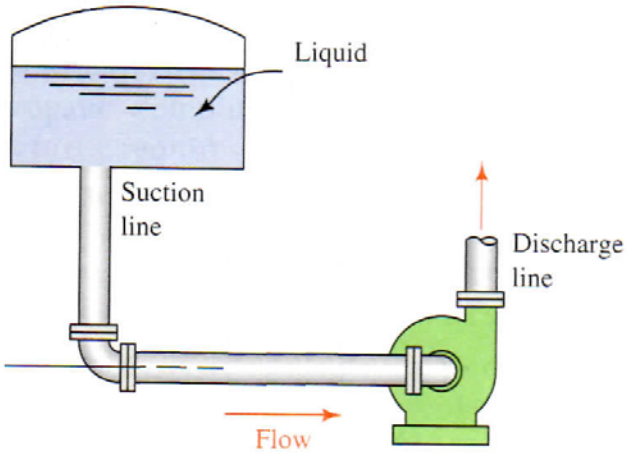
b. Write the simplified form of the Bernoulli equation for the condition where reference points 1 & 2 are chosen to be at the centreline of the piping system where the two pressure gauges are located (see figure below).
(4 marks)



c. If the upstream pressure gauge reads 692 kPa and the downstream pressure gauge reads 20 kPa, determine the volumetric flow rate of carbon tetrachloride (m^3/s) in the piping system.
(8 marks)

QUESTION 5 - 10 MARKS

The system shown in the accompanying figure (note: not drawn to scale) is designed to pump hot water (temperature of 70 ºC) from a large reservoir. The reservoir is a closed tank that is air pressurized to maintain a gauge pressure of +25 kPa. The suction line going into the pump is a 1" schedule 80 commercial steel pipe with a total length of 12 m; the elbow is a regular 90º with threaded fittings (corresponding loss coefficient K = 1.4).



a) State the correct relation between the available NPSH ($NPSH_A$) of the system and required NPSH ($NPSH_R$) of the pump that must be met to avoid cavitation.
(1 mark)

b) Determine the value of $NPSH_A$ (m) for the system given the following conditions:
- the inlet to the pump is located 2.5 m below the surface of the hot water
- the volumetric flow rate of water is 0.0017 m^3/s.

(8 marks)

c) If you are required to add a 10% margin as a safety factor, then which one(s) of the four pumps listed below would you choose to install into the system to deliver the hot water from the reservoir? **(1 mark)**

Pump A	$NPSH_R = 2.3$ m
Pump B	$NPSH_R = 2.9$ m
Pump C	$NSPH_R = 3.5$ m
Pump D	$NPSH_R = 4.1$ m

BONUS - 3 MARKS

In class, we've seen a few times the Hagen-Poiseuille equation, an analytical expression for the pressure drop for laminar flow conditions through a horizontal pipe of constant diameter.

$$\boldsymbol{\Delta P} = \frac{128\mu L}{\pi D^4} Q$$

For assignment #4, you derived the following equation for the pressure drop for turbulent flow with Re < 100,000 through a smooth, horizontal pipe of constant diameter:

$$\boldsymbol{\Delta P = 0.241 \frac{\rho^{0.75}\mu^{0.25}L}{D^{4.75}} Q^{1.75}}$$

On the pressure drop vs flow rate figure below, draw a nice sketch of the pressure drop profiles for both laminar (using the Hagen-Poiseuille equation) and turbulent (using the 2nd equation given above) flow of water at 5 °C through a smooth pipe (internal diameter = 1 cm) that is 15 m long.

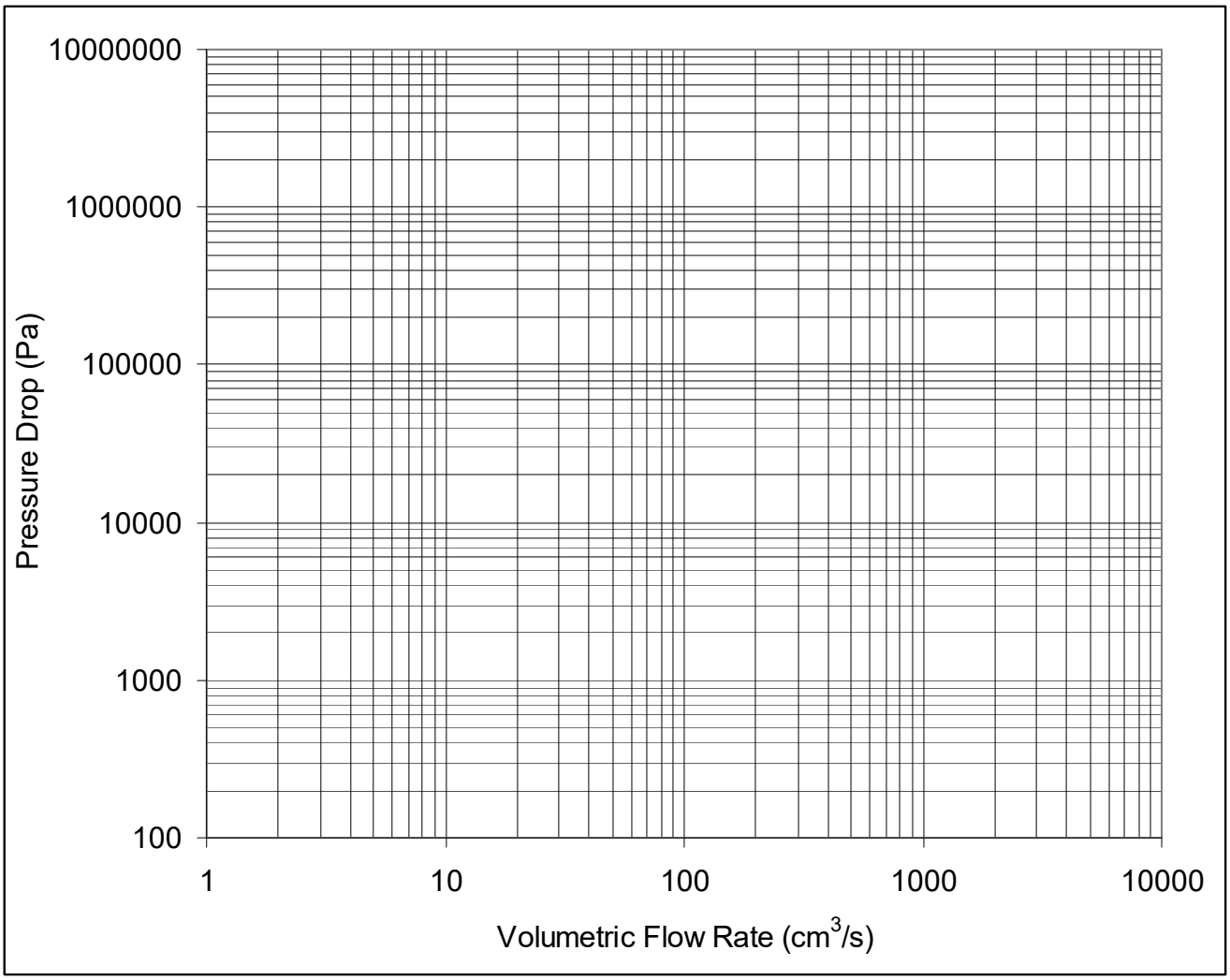


TABLE D.1. *Pipe dimensions.*

Nominal Diameter	Outside Diameter in. (ft)	Outside Diameter cm	Schedule	Inside Diameter ft	Inside Diameter cm	Flow Area ft^2	Flow Area cm^2
1/8	0.405	1.029	40 (std)	0.02242	0.683	0.0003947	0.366 4
	(0.03375)		80 (xs)	0.01792	0.547	0.0002522	0.235 0
1/4	0.540	1.372	40 (std)	0.03033	0.924	0.0007227	0.670 6
	(0.045)		80 (xs)	0.02517	0.768	0.0004974	0.463 2
3/8	0.675	1.714	40 (std)	0.04108	1.252	0.001326	1.233
	(0.05625)		80 (xs)	0.03525	1.074	0.0009759	0.905 9
1/2	0.840	2.134	40 (std)	0.05183	1.580	0.002110	1.961
	(0.070)		80 (xs)	0.04550	1.386	0.001626	1.508
			160	0.03867	1.178	0.001174	1.090
			(xxs)	0.02100	0.640	0.0003464	0.321 7
3/4	1.050	2.667	40 (std)	0.06867	2.093	0.003703	3.441
	(0.0875)		80 (xs)	0.06183	1.883	0.003003	2.785
			160	0.05100	1.555	0.002043	1.898
			(xxs)	0.03617	1.103	0.001027	0.955 5
1	1.315	3.340	40 (std)	0.08742	2.664	0.006002	5.574
	(0.1095)		80 (xs)	0.07975	2.430	0.004995	5.083
			160	0.06792	2.070	0.003623	3.365
			(xxs)	0.04992	1.522	0.001957	1.815

TABLE D.1. continued. *Pipe dimensions.*

Nominal Diameter	Outside Diameter in. (ft)	Outside Diameter cm	Schedule	Inside Diameter ft	Inside Diameter cm	Flow Area ft^2	Flow Area cm^2
2	2.375	6.034	40 (std)	0.1723	5.252	0.02330	21.66
	(0.1979)		80 (xs)	0.1616	4.926	0.02051	19.06
			160	0.1406	4.286	0.01552	14.43
			(xxs)	0.1253	3.820	0.01232	11.46
$2^1/_2$	2.875	7.303	40 (std)	0.2058	6.271	0.03325	30.89
	(0.2396)		80 (xs)	0.1936	5.901	0.02943	27.35
			160	0.1771	5.397	0.02463	22.88
			(xxs)	0.1476	4.499	0.01711	15.90
3	3.500	8.890	40 (std)	0.2557	7.792	0.05134	47.69
	(0.2917)		80 (xs)	0.2417	7.366	0.04587	42.61
			160	0.2187	6.664	0.03755	34.88
			(xxs)	0.1917	5.842	0.02885	26.80
$3^1/_2$	4.000	10.16	40 (std)	0.2957	9.012	0.06866	63.79
	(0.3333)		80 (xs)	0.2803	8.544	0.06172	57.33
4	4.500	11.43	40 (std)	0.3355	10.23	0.08841	82.19
	(0.375)		80 (xs)	0.3198	9.718	0.07984	74.17
			120	0.3020	9.204	0.07163	66.54
			160	0.2865	8.732	0.06447	59.88
			(xxs)	0.2626	8.006	0.05419	50.34
5	5.563	14.13	40 (std)	0.4206	12.82	0.1389	129.10
	(0.4636)		80 (xs)	0.4011	12.22	0.1263	117.30
			120	0.3803	11.59	0.1136	105.50
			160	0.3594	10.95	0.1015	94.17
			(xxs)	0.3386	10.32	0.09004	83.65

Table A.1

Specific Gravities of Selected Engineering Materials

(a) Common Manometer Liquids at 20°C

Liquid	Specific Gravity
E.V. Hill blue oil	0.797
Meriam red oil	0.827
Benzene	0.879
Dibutyl phthalate	1.04
Monochloronaphthalene	1.20
Carbon tetrachloride	1.595
Bromoethylbenzene (Meriam blue)	1.75
Tetrabromoethane	2.95
Mercury	13.55

Source: Data from References [1–3].

Material	Specific Gravity (—)
Aluminum	2.64
Balsa wood	0.14
Brass	8.55
Cast Iron	7.08
Concrete (cured)	2.4[a]
Concrete (liquid)	2.5[a]
Copper	8.91
Ice (0°C)	0.917
Lead	11.4
Oak	0.77
Steel	7.83
Styrofoam (1 pcf[b])	0.0160
Styrofoam (3 pcf)	0.0481
Uranium (depleted)	18.7
White pine	0.43

Source: Data from Reference [4].
[a]depending on aggregate.
[b]pounds per cubic foot.

Table A.6

Thermodynamic Properties of Common Gases at STP[a]

Gas	Chemical Symbol	Molecular Mass, M_m	R^b $\left(\frac{J}{kg \cdot K}\right)$	c_p $\left(\frac{J}{kg \cdot K}\right)$	c_v $\left(\frac{J}{kg \cdot K}\right)$	$k = \frac{c_p}{c_v}$ (–)
Air	—	28.98	286.9	1004	717.4	1.40
Carbon dioxide	CO_2	44.01	188.9	840.4	651.4	1.29
Carbon monoxide	CO	28.01	296.8	1039	742.1	1.40
Helium	He	4.003	2077	5225	3147	1.66
Hydrogen	H_2	2.016	4124	14,180	10,060	1.41
Methane	CH_4	16.04	518.3	2190	1672	1.31
Nitrogen	N_2	28.01	296.8	1039	742.0	1.40
Oxygen	O_2	32.00	259.8	909.4	649.6	1.40
Steam[c]	H_2O	18.02	461.4	~2000	~1540	~1.30

Source: Data from References [7, 16, 17].
[a]STP = standard temperature and pressure, $T = 15°C = 59°F$ and $p = 101.325$ kPa (abs) = 14.696 psia.
[b]$R \equiv R_u/M_m$; $R_u = 8314.3$ J/(kgmol · K) = 1545.3 ft · lbf/(lbmol · °R); 1 Btu = 778.2 ft · lbf.
[c]Water vapor behaves as an ideal gas when superheated by 55°C (100°F) or more.

Table A.2

Physical Properties of Common Liquids at 20°C

Liquid	Isentropic Bulk Modulus[a] (GN/m^2)	Specific Gravity (—)
Benzene	1.48	0.879
Carbon tetrachloride	1.36	1.595
Castor oil	2.11	0.969
Crude oil	—	0.82–0.92
Ethanol	—	0.789
Gasoline	—	0.72
Glycerin	4.59	1.26
Heptane	0.886	0.684
Kerosene	1.43	0.82
Lubricating oil	1.44	0.88
Methanol	—	0.796
Mercury	28.5	13.55
Octane	0.963	0.702
Seawater[b]	2.42	1.025
SAE 10W oil	—	0.92
Water	2.24	0.998

Source: Data from References [1, 5, 6].
[a]Calculated from speed of sound; 1 GN/m^2 = 10^9 N/m^2 (1 N/m^2 = 1.45×10^{-4} lbf/in.^{2}).
[b]Dynamic viscosity of seawater at 20°C is $\mu = 1.08 \times 10^{-3}$ N · s/m^2. (Thus, the kinematic viscosity of seawater is about 5 percent higher than that of freshwater.)

Table A.8

Properties of Water (SI Units)

Temperature, T (°C)	Density, ρ (kg/m^3)	Dynamic Viscosity, μ (N · s/m^2)	Kinematic Viscosity, ν (m^2/s)	Surface Tension, σ (N/m)	Vapor Pressure, p_v (kPa)	Bulk Modulus, E_v (GPa)
0	1000	1.76E-03	1.76E-06	0.0757	0.661	2.01
5	1000	1.51E-03	1.51E-06	0.0749	0.872	
10	1000	1.30E-03	1.30E-06	0.0742	1.23	
15	999	1.14E-03	1.14E-06	0.0735	1.71	
20	998	1.01E-03	1.01E-06	0.0727	2.34	2.21
25	997	8.93E-04	8.96E-07	0.0720	3.17	
30	996	8.00E-04	8.03E-07	0.0712	4.25	
35	994	7.21E-04	7.25E-07	0.0704	5.63	
40	992	6.53E-04	6.59E-07	0.0696	7.38	
45	990	5.95E-04	6.02E-07	0.0688	9.59	
50	988	5.46E-04	5.52E-07	0.0679	12.4	2.29
55	986	5.02E-04	5.09E-07	0.0671	15.8	
60	983	4.64E-04	4.72E-07	0.0662	19.9	
65	980	4.31E-04	4.40E-07	0.0654	25.0	
70	978	4.01E-04	4.10E-07	0.0645	31.2	
75	975	3.75E-04	3.85E-07	0.0636	38.6	
80	972	3.52E-04	3.62E-07	0.0627	47.4	
85	969	3.31E-04	3.41E-07	0.0618	57.8	
90	965	3.12E-04	3.23E-07	0.0608	70.1	2.12
95	962	2.95E-04	3.06E-07	0.0599	84.6	
100	958	2.79E-04	2.92E-07	0.0589	101	